HARSH KUMAR

AI Engineer — Computer Vision | LLMs & RAG | Real-Time Systems
9720396468 | harshji9756@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building real-time computer vision and GenAI systems, including a violence-detection pipeline (RT-DETR + VideoMAE) at **~26 FPS** on live RTSP feeds, a license-plate recognition system trained on **14K+ images (96% mAP@50)**, and a **production voice-cloning SaaS** serving live traffic. Owns projects end-to-end: model training, pipeline engineering, and FastAPI deployment. **AWS Certified**; **SIH internal-round winner (2023)**.

TECHNICAL SKILLS

Computer Vision: YOLOv11/RT-DETR (Ultralytics), VideoMAE, OpenCV, ByteTrack, PaddleOCR, OmniParser
LLMs & GenAI: LangChain, HuggingFace Transformers, Qwen2.5-VL, OpenAI API, RAG, Vector Databases, Whisper
Audio/Speech: Coqui XTTS v2 (voice cloning), spaCy (NLP preprocessing)
Frameworks: PyTorch, TensorFlow, Scikit-learn
Backend & Serving: FastAPI, Docker, Redis, PostgreSQL, MongoDB (incl. GridFS)
Cloud & Tools: AWS (EC2, S3), Git, Linux, Prometheus
Data: SQL, Pandas, NumPy, Vector DBs

PROFESSIONAL EXPERIENCE

AI Engineer | SysTools Software Pvt Ltd *Oct 2025 – Present*

- Built and deployed a real-time violence/crowd detection system (RT-DETR person tracking + fine-tuned VideoMAE classifier) processing live RTSP surveillance feeds at **~26 FPS**, using a multi-factor heuristic suspicion gate (proximity, motion, optical-flow chaos) to limit costly transformer inference to only high-risk events.
- Engineered a forensic email-security pipeline validating SPF/DKIM/DMARC/ARC authentication and performing HTML/URL/attachment forensic analysis with a weighted rule-based risk-scoring engine; served via FastAPI with MongoDB persistence, averaging **under 1 second** per email scan.
- Built and deployed a production voice-cloning SaaS (Coqui XTTS v2, FastAPI, Redis + MongoDB GridFS 2-tier caching) live at a public URL supporting **18 speaker profiles** across English and Hindi; caching cut repeat-request latency from **~60s to ~2s**.

Machine Learning Intern | CodSoft *Jun 2024 – Jul 2024*

- Built and evaluated classical ML models (Titanic survival, Iris classification, House Price regression) through feature engineering and hyperparameter tuning, achieving **85% and 96% accuracy** respectively on the first two tasks.

KEY PROJECTS

- **AI Screenshot-to-Video Automation —** 8-stage FastAPI pipeline (PaddleOCR + OmniParser/YOLO+Florence-2 UI detection, two-pass Qwen2.5-VL-32B/7B narration, Whisper-aligned subtitles, FFmpeg rendering) converting static UI screenshots into narrated 1080p walkthrough videos in **~90–110s end-to-end**; engineered sequential 4-bit model loading/unloading to fit 32B+7B models within limited GPU VRAM. Live Demo
- **Fight, Riot & Crowd Detection —** Real-time violence-detection pipeline combining RT-DETR tracking with a fine-tuned 3-class VideoMAE transformer, gated by a custom suspicion score (spatial clustering, IoU, optical-flow motion/chaos) to avoid running the transformer on every frame — sustaining **~26 FPS** on live RTSP feeds while auto-adapting between CCTV and mobile camera profiles.
- **ANPR — Automatic Number Plate Recognition (Trinetra-Parity) —** Multi-threaded ANPR pipeline (YOLOv11 detection + dual-ensemble PaddleOCR) for real-time plate recognition over RTSP/HLS; fine-tuned a plate detector on **14,142 labeled images to 96.1% mAP@50 / 93.3% recall**, and an OCR recognizer on **9,425 samples to 71% exact-match accuracy** at 16–36ms per-plate CPU inference. Added multi-frame consensus voting and an active-learning review dashboard that feeds corrected reads back into training.
- **Voice Cloning Engine (XTTS) —** End-to-end TTS/voice-cloning service (Coqui XTTS v2) with a linguistic preprocessing pipeline (spaCy POS-based homograph disambiguation, Hindi numeral/punctuation normalization) and a 2-tier Redis + MongoDB GridFS cache, streaming audio via FastAPI to a live public-facing app. Live Demo
- **RAG-Based Search Engine —** Retrieval-Augmented Generation system using LangChain, embeddings, and a vector database to enable context-aware semantic search over a document corpus, combining retrieval with LLM-generated answers.
- **Email Threat Forensics Engine (rule-based) —** Forensic email-security system validating SPF/DKIM/DMARC/ARC authentication and HTML/URL/brand-impersonation signals through a weighted rule-based risk engine; classifies emails into Phishing/Malware/Scam/Compromised/Safe and exports structured JSON/Excel/CSV reports. Live Demo

EDUCATION

B.Tech — Artificial Intelligence & Machine Learning *2021 – 2025*
Senior Secondary (12th) — Science (PCM) 2020

ACHIEVEMENTS & CERTIFICATIONS

- AWS Cloud Practitioner Essentials — AWS Training & Certification (2026)
- Smart India Hackathon (SIH) — Internal Round Winner (2023)
- IBM Python 101 for Data Science Certificate (2022) — IBM